Matched logarithmic source Analytic supplement

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Scope and applicability

This companion expands the finite logarithmic and grouped-head estimates used in *Retuned translated probes and the conditional bound* $\Lambda < 0.116398$ *for the de Bruijn–Newman constant*, version 1.0. The statements below are source estimates at the prescribed rational heights, not extrapolations from a comparison trajectory.

The retuned production uses `evidence/PLAN_NEW.json`: all 608 cells were regenerated in two interval constructions, including the 274 cells whose heights changed. The final main paper uses the complete selected source `evidence/SOURCE_SELECTED.json`; a frozen cellwise selector chooses between valid original and new source rows. An estimate is reused only at exactly the same time and height. The full matched-source algorithms and original analytic text remain in the immutable evidence.

Sections 1–8 give formal positivity, matched coefficient accounting, the grouped-head alternative and complete displacement-band estimates. The scalar endpoints of the historical source are omitted because they are not inputs to the new theorem. The sharp-mode explanation in Section 8 uses the corrected pointwise displacement and full-band argument. The two current main histories use fixed shifts only, so that continuum mode contributes no step to the headline.

The separately supplied *Prior Analytic Foundations*, Part A, gives the enlarged-height approximation, both reflected branches, cutoff-activation estimates, denominator exclusion and Cauchy derivative transfer. Its Gaussian majorant is replaced by the corrected all-height bound in Appendix B of the main paper. The same correction applies to both original and retuned source records. The narrower historical wall error is not used. Part B of the retained Foundations contains the older kernel catalogue; the current main paper proves and lists the 44 new certificates separately.

The 7/8 zero-free hypothesis remains an assumption. The source calculation does not depend on the historical scalar endpoint, a narrower arithmetic strip, or a later finite-frequency phase certificate. The accompanying programs check the finite enclosures and their reconstruction; the written analytic arguments retain a separate mathematical review obligation.

Contents

Scope and applicability **1**

1. Function, global inputs and source schedule **3**

2. Original one-point physical transfer **3**

3. Formal positive coefficients **4**

4. Cancelling the constant overlap before making enclosures **4**

5. Complete translated logarithmic deficit **5**

6. Grouped head–head surplus, kept as a separate alternative **6**

7. How the finite arithmetic is enclosed **6**

8. Matching the phase enclosure to the current kernel parameter **6**

Proof-to-code map for the present release **7**

Mathematical references **8**

1. Function, global inputs and source schedule

Use the original normalisation

$$H_t(z) = \int_0^\infty e^{tu^2}\Phi(u)\cos(zu)\,du, \qquad H_0(z) = \xi((1+iz)/2)/8, \qquad \partial_t H_t = -\partial_z^2 H_t.$$

Set $X_0 = 6\,000\,000\,185\,827$ and $N_0 = 690\,988$. Each source configuration (original or retuned) gives 608 closed time rows $[i/4000, (i+1)/4000]$, each with a prescribed rational height h_i. All heights lie in the parameter domain of the retained larger-height derivation. Every scalar error condition is evaluated at the prescribed height. The unchanged program `certify_coupled_probe.py` produces the finite inequalities from this rational configuration; the source floors are not inferred from an earlier comparison trajectory.

The six frequency ranges have boundaries

$$\log N_0 + 0, \quad \log N_0 + 1/20, \quad \log N_0 + 3/20, \quad \log N_0 + 2/5, \quad \log N_0 + 1, \quad \log N_0 + 2, \quad +\infty.$$

The original activation estimates cover every native cutoff, including all future cutoffs in the infinite final range. They are re-evaluated, not replaced by a finite physical-frequency sample. At two probe points $x \pm a$, their possible source-frequency classes are the original eleven pairs (j, j) and $(j, j+1)$; the original separation argument still applies because every prescribed displacement has magnitude below two and the frequency intervals are unchanged.

The finite-head wall and localisation/confinement conditions retain the same domain. Both numerical wall constructions were regenerated at 256 and 384 bits; their mathematical interval records agree with the retained originals. The physical wall conclusion is interpreted using the current paper's corrected broad Gaussian debit, not the narrower historical `physical_eC0` comparison. The published finite-RH verification itself is imported, not rerun. Their dependency and actual execution scope are stated in `RESULT.json`. The exact comparison consumer checks the complete source before accepting a conditional endpoint.

2. Original one-point physical transfer

Write $S(x,h,t) = -\Im(H_t'/H_t)(x+ih)$. The leading source uses

$$s_U^0 = (1+h)/2 + tL/4 - i(x/2 + \pi t/8), \qquad L = \log(x/(4\pi)),$$

and the same reflected branch and physical carrier as the parent. Let

$$E_t(s) = \prod_{p\in\{2,3,5\}} (1 - b_t(p)p^{-s}), \qquad b_t(n) = e^{t\log^2 n/4},$$

and let Q be the complete multiplied finite source. Its labelled representation and height derivative are

$$Q = 1 + \sum_{j>0} a_j z_j, \qquad V = \sum_j \lambda_j a_j z_j, \quad Q_h = -V/2,$$

with $a_j \geq 0$, $|z_j| = 1$ and $\lambda_j \geq 0$. All reflected and product-tail labels are retained. The reassembled bounds $\bar{A} \geq \sum_{j>0} a_j$ and $\bar{B} \geq \sum_j \lambda_j a_j$ satisfy $\bar{A} < 4/5$. The one-point cone constant C and its physical floor ℓ give the transfer

$$S > \ell + \frac{C - \Re(V/Q)}{2} + (E_\ell - \Re(E_h/E))\,. \tag{2.1}$$

The full value and derivative approximation errors are already subtracted in ℓ. This identity does not differentiate a centre-only remainder or assume equal denominators at the two source points.

3. Formal positive coefficients

All algebra in this section is formal Dirichlet convolution, not convergence of an infinite positive-time Dirichlet series. Write $Df(n) = (\log n)f(n)$, $c_t = b_t * \prod_{p\in\{2,3,5\}}(\delta - b_t(p)\delta_p)$, and

$$\widetilde{d}_t = D\log_* c_t.$$

For every n and $t \geq 0$, the coefficients $c_t(n)$ and $\widetilde{d}_t(n)$ are nonnegative and nondecreasing. In fact every time derivative is nonnegative. Here is a self-contained proof of the needed monotonicity.

For $d_t = D\log_* b_t$, the derivation rule and $\partial_t b_t = D^2 b_t/4$ give

$$\partial_t d_t(n) = \frac{\log^2 n}{4} d_t(n) + \frac{\log n}{4} \sum_{ab=n,\ a,b\geq 2} d_t(a)d_t(b). \tag{3.1}$$

The initial values are the nonnegative von Mangoldt coefficients. Induction on n and the integrating-factor formula prove positivity of every time derivative. For a selected prime p and $n = p^k$,

$$\widetilde{d}_t(p^k) = d_t(p^k) - (\log p)e^{kt\log^2 p/4}.$$

Writing $\kappa = (\log p)^2/4$, this difference satisfies

$$e_k' = k^2\kappa e_k + (k^2-k)\kappa(\log p)e^{k\kappa t} + \frac{k\log p}{4}\sum_{j=1}^{k-1} d_t(p^j)d_t(p^{k-j}), \quad e_k(0) = 0.$$

The forcing and all its derivatives are nonnegative. This proves the assertion for $\widetilde{d}_t$, including exact removal of the selected primes. Finally

$$(\log n)c_t(n) = \sum_{d|n,\ d\geq 2} \widetilde{d}_t(d)c_t(n/d) \tag{3.2}$$

proves the same assertion for c_t by induction. In particular, $c_t(1) = 1$ and $c_t(p) = \widetilde{d}_t(p) = 0$ for $p \in \{2,3,5\}$.

The positive-time infinite numerical series with coefficients $b_t(n)$ converges nowhere at a fixed abscissa. Nothing below uses it as an analytic function.

4. Cancelling the constant overlap before making enclosures

Let the complete first head Q_0 run through N_0, and select $K = 8192$. At fixed t and $\sigma = \Re(s_U^0)$, let

$$T = \sum_{2\leq d\leq K} \widetilde{d}_t(d)d^{-\sigma}z_d, \quad D_K = \sum_{2\leq d\leq K} \widetilde{d}_t(d)d^{-\sigma}.$$

The physical first-head phases obey $z_d z_m = z_{dm}$. This is used only in that head; no compatible phase is imposed on the reflected or other tail labels.

Define the **constant-removed common overlap**

$$Z = \sum_{2\leq d\leq K} \widetilde{d}_t(d)d^{-\sigma} \sum_{2\leq m\leq N_0/d} c_t(m)m^{-\sigma}. \tag{4.1}$$

Every summand is nonnegative. The total matched coefficient mass between QT and V through N_0 is $D_K + Z$, by (3.2). Removing that mass from both labelled sums gives

$$|QT - V| \leq \bar{B} + (1+\bar{A})D_K - 2(D_K + Z) = \bar{B} - (1-\bar{A})D_K - 2Z. \tag{4.2}$$

The right side is nonnegative, since it is an upper bound for a residual after legitimate matching (and the use of upper mass bounds only increases it). Since $|Q| \geq 1 - \bar{A}$,

$$\boxed{\Re(V/Q) \leq \frac{\bar{B} - 2Z}{1 - \bar{A}} - (D_K - \Re T).} \tag{4.3}$$

This is just exact cancellation inside the finite quotient estimate, not a new global nonvanishing principle. Its numerical advantage is specific: the constant contribution D_K is **not** bounded above in one location and below in another. The only remaining overlap Z is a positive sum and can be bounded directly from below on the complete parameter rectangle.

Substitution into (2.1) gives

$$S > \ell + C/2 - \frac{\bar{B} - 2Z_{\mathrm{low}}}{2(1 - \bar{A})} + (D_K - \Re T)/2 + (E_\ell - \Re(E_h/E)). \tag{4.4}$$

All old approximation errors, full masses, and denominator conditions remain paid. The accepted bound does not add (4.4) to the old cone-surplus bound; those are alternative lower bounds for the same physical quantity.

5. Complete translated logarithmic deficit

At $x \pm a$, for each selected index,

$$z_d^{\pm} = z_d^c e^{\pm ia \log d/2}.$$

For any nonnegative coefficients $q_d^{\pm} \geq q_d^{\mathrm{low}}$,

$$\frac{q_d^+(1 - \Re z_d^+) + q_d^-(1 - \Re z_d^-)}{2} \geq q_d^{\mathrm{low}} \left(1 - |\cos(a \log d/2)|\right). \tag{5.1}$$

Thus, if the points belong to source rectangles $j \leq k$, the lower coefficient from the higher upper abscissa $\sigma_{k,+}$ is safe at both. The positive logarithmic deficit lower bound is

$$G_{k,a} = \sum_{2 \leq d \leq K} \widetilde{d}_{t_-}(d) d^{-\sigma_{k,+}} \inf_{a \in I_a} \left(1 - |\cos(a \log d/2)|\right), \quad \sigma_{k,+} = (1 + h)/2 + t_+ L_{k,+}/4. \tag{5.2}$$

Coefficient monotonicity from Section 3 and monotonicity of $n^{-\sigma}$ justify these endpoints. The complete displacement interval is retained for the sharp continuum kernel, not replaced by its value at a floating comparison point.

Let $F_j = \ell_j + C_j/2 - (\bar{B}_j - 2Z_{\mathrm{low},j})/[2(1 - \bar{A}_j)]$. The new lower bound for the physical paired source is

$$\boxed{\tfrac{1}{2}[S(x - a, h, t) + S(x + a, h, t)] > (F_j + F_k + G_{k,a})/2 + G_{jk,a}^{\mathrm{Euler}}.} \tag{5.3}$$

The Euler term is the independently valid existing paired Euler deficit, not a newly assumed Euler product for the heat-flowed field.

The initial version of this new source is produced only for the first 320 time rows and the first five, bounded frequency classes, on the five named displacement modes in the configuration. Section 8 adds the other six catalogue modes and the sixteen sharp-parameter bands within the same stated time range. Elsewhere the complete original paired source remains available. A pointwise maximum of valid alternatives on each source-class pair is valid. The global floor is the minimum of those accepted pair-specific floors over **all eleven** possible pairs.

6. Grouped head–head surplus, kept as a separate alternative

The exact original surplus identity is

$$C|Q|^2 - \Re(V\bar{Q}) = M_C + \sum_{w_{mn}\geq 0} w_{mn}(1 - \Re(z_m\bar{z}_n)) + \sum_{w_{mn}<0} |w_{mn}|(1 + \Re(z_m\bar{z}_n)),$$

where $w_{mn} = a_m a_n(\log m + \log n - 2C)$ on the first head and $M_C \geq 0$ is proved by the full original negative-pair budget. The original correlated producer selects only constant–head pairs. The present additional selection uses disjoint ordered groups of indices in $\{2, \ldots, 2048\}$, so it does not double-count any of those pairs.

For two groups $G < H$, bound $\log m + \log n$ below and above by their complete endpoint sums. If those bounds do not fix a common sign for both C_j and C_k, take zero credit. Otherwise take the smaller signed distance to those two thresholds. Bound the phase argument $a(\log n - \log m)/2$ on its whole interval; an interval containing an integer multiple of π gets zero credit. On the remaining intervals the maximum absolute cosine is attained at an endpoint.

Multiply this gap and attenuation by the two lower group masses from the higher upper abscissa and divide by $2(1 + \max(\bar{A}_j, \bar{A}_k))^2$. The result is a valid additional paired cone-surplus floor. For distinct groups, the double coefficient sum factors exactly into the product of the two group masses. Diagonal within-group terms are omitted as nonnegative discarded surplus.

Only source pairs $(0, 0)$, $(0, 1)$ and $(1, 1)$ receive this extra credit; all other class pairs keep their existing bounds. The credit is combined with the original cone bound. It is **not added** to the logarithmic alternative (5.3); the exact consumer selects the maximum of the two separately proved alternatives.

7. How the finite arithmetic is enclosed

At each lower time endpoint the producer evaluates the complete c_t coefficients through K for the logarithmic rows, and through 2048 for the remaining grouped-surplus rows. The two lanes construct the finite Euler deflation in different orders. Logarithmic coefficients are generated by two forms of the finite recurrence (3.2), one forward subtractive convolution and one divisor-by-divisor reverse accumulation.

To evaluate (4.1), no head truncation below N_0 is silently introduced. For every selected d, all $m \leq \lfloor N_0/d \rfloor$ are retained. Positive raw heat weights are enclosed by endpoint bins with every required $\lfloor N_0/(de) \rfloor$, $e \mid 30$, inserted as a bin boundary. At $\sigma_{j,+}$ these raw weights are decreasing; the producer explicitly checks the derivative condition. Exact finite Euler inclusion–exclusion gives the full c_t-prefix. Its constant term one is subtracted before the lower bound is taken. The proven positivity of the remaining prefix permits replacement of a negative enclosure lower endpoint by zero.

The complete overlap, logarithmic deficits, group masses and cosine attenuations are rounded down. Their arithmetic combinations and all source-floor selection use exact integers and rational numbers. No floating number from the discovery plan determines an acceptance inequality. The rational PLAN is an input to be checked, not a certificate of its own success.

The logarithmic lower grids are regenerated and hashed; their scalar contractions are recorded. The grouped-head masses and complete group-attenuation grids are retained for rational reconstruction. Both transcendental lanes use Arb/FLINT. Their agreement is not independent-library verification.

8. Matching the phase enclosure to the current kernel parameter

The original sharp-source estimate encloses its displacement over the whole range

$$a \in [h\sqrt{(1 - 51/250)/3},\ h/\sqrt{3}].$$

That estimate is valid. A dynamic comparison can use a narrower displacement interval when that entire interval is certified. A separate, freshly regenerated supplier partitions the dimensionless parameter into the **sixteen complete closed bands**

$$I_b = [51b/4000,\ 51(b+1)/4000], \qquad b = 0, \ldots, 15.$$

For each band and prescribed time row, it evaluates all selected phase deficits throughout

$$a \in [h\sqrt{(1-51(b+1)/4000)/3},\ h\sqrt{(1-51b/4000)/3}].$$

The square-root endpoints are rounded outwards on the exact 80-bit grid. The interval cosine is then evaluated on the entire resulting phase interval. Thus `max(0, floor_grid(1-abs(cos(interval))))` is a lower bound for the complete nonnegative phase deficit, including any trigonometric extremum inside the interval. No centre sample or monotonicity of the cosine is assumed.

At every time row, the original complete constant-head and eight-harmonic Euler gains are regenerated for all sixteen bands and all eleven physical source-class pairs. During the first 320 rows, the matched logarithmic deficit through 8192 is also regenerated; its entire coefficient grid must agree exactly with the corresponding earlier arithmetic construction. The previously certified complete Z lower bound is reused without change, not estimated at a floating comparison point. The same supplier adds logarithmic alternatives for the six catalogue displacement ratios absent from the first finite-log supplier. All other portions retain their complete earlier source bounds.

The exact source consumer takes a maximum of separately valid alternatives on each physical source-class pair and then the minimum over all eleven pairs. It records one complete source floor per dimensionless band. The older broad sharp floor remains an alternative, so no previously established floor is lost.

Corrected use of the band source. At the beginning of a rational substep, let $v_{\mathrm{in}} = q_{\mathrm{in}}/h^2$. At a contact with the comparison at squared height $Q(t)$, put $w = Q(t)/h^2$ and choose $a(w) = h\sqrt{(1-w)/3}$. The physical source must cover every displacement in the interval swept as w decreases during that substep. Take the minimum of its certified floors over all such complete bands. The own-pair cap is $6/(4-w) \le 6/(4-v_{\mathrm{in}})$. No derivative of a moving probe is taken. The affine tangent $q_{\mathrm{in}} - (2 + cq_{\mathrm{in}})\Delta t$ lies below the exact comparison throughout the step and therefore supplies a conservative lower endpoint for the swept height interval. The independent auditor recomputes it from the final rate and checks every required band.

This paragraph explicitly **replaces**, rather than endorses, the historical fixed-displacement explanation. For $v_{\mathrm{in}} = 1/5$, the incoming sharp choice has $r^2 = 4/15$ and self cap $30/19$. Holding that shift fixed at $w = 1/10$ gives self contribution $2100/1321 = 30/19 + 270/25099$, so the historical assertion fails. The source's complete displacement-interval bounds are unaffected. The calculation above gives the correction. Section 6 of the current main paper explains why its fixed-shift histories do not use this mode.

This refinement is independent of the choice of the discovery curve: a floating trajectory proposes the rational height schedule, while each band's full source statement is proved before the exact trajectory is integrated.

Proof-to-code map for the present release

The retained matched-source algorithms are unchanged. Their inputs and complete output rows are reconstructed at the original and retuned plans before the final source selection is accepted.

Mathematical component	Implementation in the evidence archive
Physical enclosures at the prescribed heights	Original `certify_coupled_probe.py`; exact parent-row validation
Constant/head and Euler gains	Retained `certify_joint.py`
Positive logarithmic coefficients and grouped-head alternatives	Retained `certify_arithmetic.py`
Complete paired lower bounds	`forward_common.parent_rows` and `assemble_forward.build`
Original focused displacement bands	Retained `certify_focused.py` and `assemble_focused.construct`; not rerun at changed heights
Retuned source and frozen whole-cell choice	`code/rebuild_source.py` at the outer evidence root

Mathematical component	Implementation in the evidence archive
New kernel certificates	Outer `code/verify_geometry.py`
Gaussian and physical-error safety	Baseline `code/check_analytic_safety.py`, applied to both sources
Exact histories and separate scalar checks	Baseline `code/produce_history.py` and `code/audit_history.py`, called with each new configuration

For each source-class pair, the valid alternatives are maximised; the minimum is then taken over the eleven complete pairs. The retuned source uses a broad valid continuum bound in each narrow band. At identical original and new time–height cells, stronger original focused estimates may be retained. No narrow-band estimate is transferred to a changed height. The main histories use the two fixed translations only.

At the root of the extracted evidence, `RUN.py -audit -output DIR` reconstructs the complete stored source and checks all eight final-source histories. Adding `-regenerate-histories` produces those histories again and requires byte identity. The separate `code/replay_native.py` wrapper reruns the retuned transcendental producers in the pinned numerical environment. It retains the old walls and does not run the focused-band producer at changed heights. The *Verification Guide* gives exact commands and identifies the output receipts.

Mathematical references

The principal external analytic framework is D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann xi function, and a new upper bound for the de Bruijn–Newman constant*, arXiv:1904.12438v2, especially Proposition 3.1, Theorem 3.2, Theorem 1.5 and Section 6. The finite localisation input is D. J. Platt and T. S. Trudgian, *The Riemann hypothesis is true up to* $3 \cdot 10^{12}$, arXiv:2004.09765, Theorem 1 with exact height $3\,000\,175\,332\,800$.

The author's prior public source is doi:10.5281/zenodo.22867547. The original matched source is the 6 October 2026 package `BCLD_MATCHED_LOG_FORWARD_20261006`. Its source-only components are preserved in this new deposit. The current main paper supplies the global fixed-shift contact proof and the new conditional theorem; it is authoritative over the historical scalar explanation.